THE RESULTS OF
YOUR ANALYSIS

PREPARED FOR

Velva Williams

DATE OF INJURY : 9/7/2019
DATE OF ANALYSIS : 9/20/2019
DATE OF IMAGES : 9/19/2019

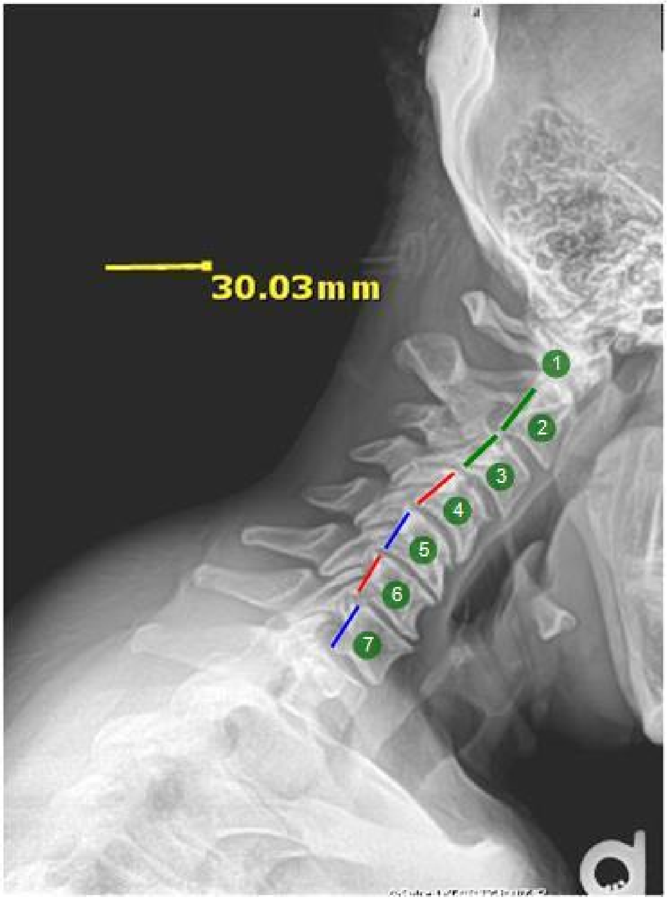
This report contains important information concerning structural changes that can be affecting your overall level of health and well-being. Spinal stability is a basic requirement for the protection of your nervous structures and the prevention of early mechanical deterioration of your spinal component.

Instability is generally considered to be a global increase in the movements associated with the occurrence of back, neck, and/or nerve root pain. Damage to any spinal structure produces some degree of spinal instability.

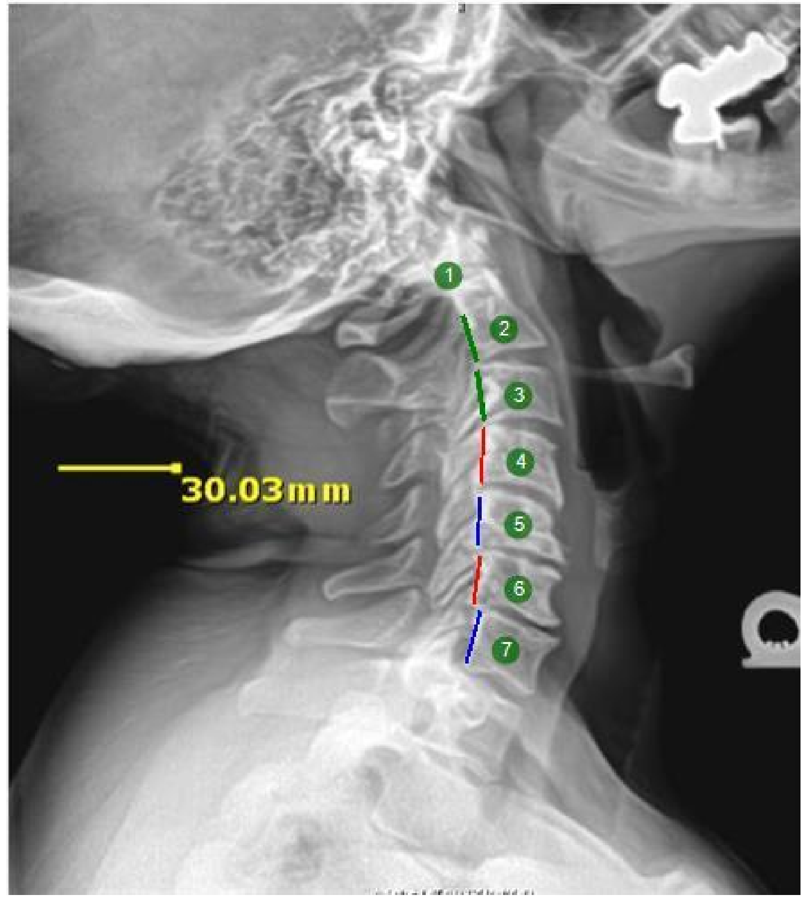
This computer aided digital analysis is an overview of your current level of spinal stability.

Cervical Flexion / Extension

PATIENT RESULT



PATIENT RESULT



RED denotes the location of the posterior longitudinal ligament (PLL) and surpasses normal acceptable segmental motion indicating the location of posterior longitudinal ligament laxity.

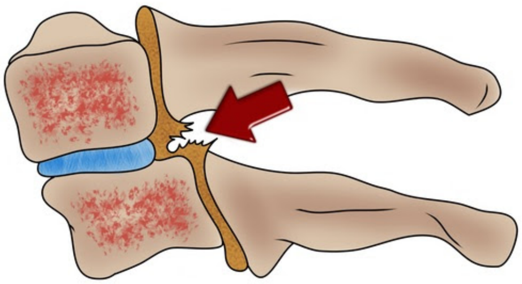
GREEN denotes the location of the posterior longitudinal ligament (PLL) with normal alignment to the surrounding vertebrae indicating no significant ligamentous laxity.

BLUE denotes a segmental motion that is clinically significant.

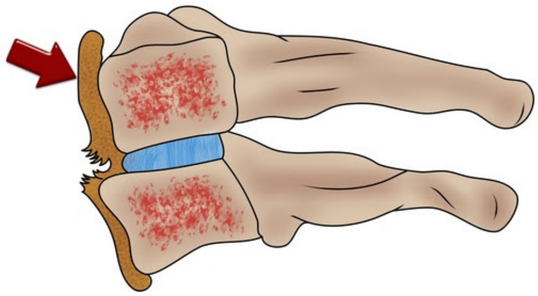
Alteration of motion segment integrity is defined as abnormal translation or angular motion between two adjacent vertebrae, traumatically induced. This motion is measured using lateral flexion and extension x-rays. Translational motion segment integrity is assessed on flexion and extension imaging by comparing and measuring the position of the posterior edge of adjacent vertebral bodies in each position. Alteration of angular motion segment integrity is assessed by comparing the angles formed by the superior endplates of adjacent vertebral bodies (adjacent motion segments). These spinal translation and angulation are measured using a process referred to as Digital Radiographic Mensuration Analysis (DRMA). SpineTech Pro software captures articular motion to quantify spinal motion pathology and identify the presence and location of ligamentous laxity (ICD-9 728.4 and ICD-10 M24.28). The World Health Organization has rated this method of analysis as "ESTABLISHED".

Cervical Flexion / Extension - Translational Motion Results

Posterior Longitudinal Ligament Injury



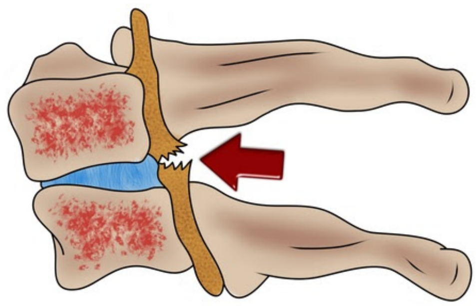
Anterior Longitudinal Ligament Injury



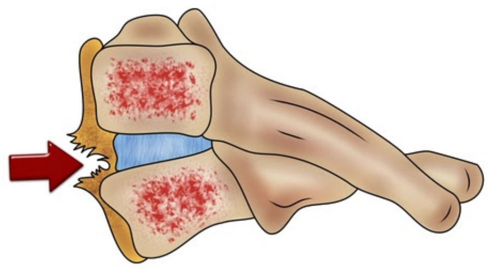
Vertebral Level	Flexion	Extension	Patient Result	Clinically Significant	Rateable Limit	Comments
C2-C3	0.37	0.1	0.47 mm	1-3.5 mm	> 3.5 mm	Normal
C3-C4	0.22	0.45	0.67 mm	1-3.5 mm	> 3.5 mm	Normal
C4-C5	0.07	0.93	1 mm	1-3.5 mm	> 3.5 mm	Normal
C5-C6	0.21	1.5	1.71 mm	1-3.5 mm	> 3.5 mm	Clinically Significant
C6-C7	2.53	2.24	4.77 mm	1-3.5 mm	> 3.5 mm	Loss of Motion Segment Integrity

Cervical Flexion / Extension - Angular Motion Results

FLEXION Posterior Longitudinal Ligament Injury



EXTENSION Anterior Longitudinal Ligament Injury



Vertebral Level	Flexion	Clinically Significant	Rateable Limit	Comments
C2-C3	-8.59	7-11°	> 11°	Normal
C3-C4	-3	7-11°	> 11°	Normal
C4-C5	16.61	7-11°	> 11°	Loss of Motion Segment Integrity
C5-C6	2.06	7-11°	> 11°	Normal
C6-C7	-2.15	7-11°	> 11°	Normal

Alteration Of Motion Segment Integrity (AOMSI) Analysis

PATIENT'S NAME: Velva Williams
DATE OF BIRTH: 08/06/1965
DATE OF INJURY: 09/07/2019
DATE OF ANALYSIS: 09/20/2019

Digital Radiographic Mensuration Analysis (DRMA)

DRMA analysis is a digital radiologic analysis necessary to determine AOMSI and alignment and ligamentous stability after trauma. DRMA technology provides an accurate diagnosis. In addition, DRMA follows the strict criteria of the **AMA Guides to the Evaluation of Permanent Impairment,** 5th Edition in calculating impairment from lateral Cervical Flexion / Extension flexion and extension radiographs. (Cocchiarella Land Andersson GBJ. AMA Press 2001). This is not a radiology report, it is a distinct and separate biomechanical analysis and the findings should be clinically correlated by the treating physician.

Lateral Flexion/Extension Views of the Cervical Flexion / Extension Spine Were Obtained and Reviewed for AOMSI:

- The angular motion segment integrity is ratable for permanent impairment at C4-C5 at 16.61°.
- The translation motion segment integrity is ratable for permanent impairment at C6-C7 at 4.77 mm but is clinically significant at C5-C6 at 1.71 mm.

IMPRESSIONS :

- Ligamentous instability is indicated in the Cervical Flexion / Extension spine. The patient has tested positive for one of the highest rated spinal injury types as recognized by strict criteria of the AMA Guides for diagnosing this injury referred to as "Alteration of Motion segment integrity"(AOMSI). These findings must be clinically correlated with the doctor’s clinical findings.
- This patient's Cervical Flexion / Extension analysis indicates Cervical Flexion / Extension Translation Motion Segment Integrity abnormality(AOMSI) and is a ratable loss. According to the AMA Guides, Fifth Edition this loss of motion segment integrity is ratable at 25%. An abnormal measurement of more than 3.5 mm of Translation Variation at one level results in instability and equates to a whole person impairment of 25%. Ligamentous instability(AOMSI) is present at C6-C7 at 4.77 mm.
- Translation variation of Motion Segment Integrity is clinically significant, at C5-C6 at 1.71 mm and this may add to this patient's total impairment.
- This patient's Cervical Flexion / Extension analysis indicates Cervical Flexion / Extension Angular Motion Segment Integrity abnormality(AOMSI) and is a ratable loss. According to the AMA Guides, Fifth Edition this loss of motion segment integrity is ratable at 25%. An abnormal measurement of more than 11° of Angular Variation at one level results in instability and equates to a whole person impairment of 25%. Ligamentous instability(AOMSI) is present at C4-C5 at 16.61°.

Dr. David Sample

Analysis Notes:

Measuring Translation :

In any of the spinal segments, lateral flexion and extension x-rays are analyzed. In the presence of true loss of motion segment integrity, the upper vertebrae will move forward on the lower vertebrae in flexion and move backwards on the lower vertebra in extension. Thus, the line along the back cortex of the upper vertebra will be anterior to the line along the posterior cortex of the lower vertebra on the flexion x-ray and posterior to it on the extension x-ray. The measurable angle is the sum of the two measurements.
If the total of the of the two measurement exceeds 4.5 mm in the lumbar spine or 3.5 mm in the cervical spine, the definition of loss of motion segment integrity is met. In the event that the line along the posterior cortex of the upper vertebra remains anterior to the line along the posterior cortex of the lower vertebra on the extension view, then consider only the distance between the lines on the flexion view, as it would not be appropriate to subtract the distance on the extension view from that on the flexion view.

Measuring Angular Motion:

Lateral flexion and extension x-rays are used. Lines are drawn along the superior cortex of the two vertebrae adjacent to the level in question on both the flexion and the extension xrays and the angles where these lines intersect measured on both views. The angle on the extension view is then subtracted from the angle on the flexion view. Normally, the angle on the flexion view will be positive and the angle on the extension view will be negative. The net result will be that the negative extension angle will be subtracted from the positive flexion angle resulting in a net addition of the two angles. (+8 –(-18) = +26). If the angle is greater than 15 degrees at L1-2, L2-3 or L3-4 or greater than 20 degrees at L4-5 or greater than 25 degrees at L5-S1, the definition of loss of motion segment integrity is met.

In the cervical spine, the method is different. A lateral flexion view only is taken. Lines are drawn along the inferior cortex of the two vertebrae above and below the level in question. Three angles are measured, the angles at the levels above and below the level in question and the angle at the level in question. If the flexion angle at the level in question is more than 11 degrees greater than either of the angles at the levels above or below, the definition of loss of motion segment integrity is met. The figure in the 5th Edition is correct, but the text under the figure suggests that both flexion and extension films are necessary when actually only a flexion view is necessary.

These are the only definitions of loss of motion segment integrity.